ITA0608 - Machine Learning
List of Lab Exercises

9.Compare Linear and Polynomial Regression using Python.

Aim

To compare the performance of **Linear Regression** and **Polynomial Regression** using Python.

Algorithm

1. Import the required Python libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Train the Linear Regression model.
5. Transform the data into polynomial features.
6. Train the Polynomial Regression model.
7. Predict the test data using both models.
8. Compare their R^2 scores and display the results.

Program Code

```
from sklearn.datasets import load_diabetes

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.preprocessing import PolynomialFeatures


data = load_diabetes()

X = data.data

y = data.target
```

```python
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

linear_model = LinearRegression()
linear_model.fit(X_train, y_train)
linear_score = linear_model.score(X_test, y_test)

poly = PolynomialFeatures(degree=2)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)

poly_model = LinearRegression()
poly_model.fit(X_train_poly, y_train)
poly_score = poly_model.score(X_test_poly, y_test)

print("Linear Regression R2 Score:", linear_score)
print("Polynomial Regression R2 Score:", poly_score)
```

Output

```
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures

data = load_diabetes()
X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

linear_model = LinearRegression()
linear_model.fit(X_train, y_train)
linear_score = linear_model.score(X_test, y_test)

poly = PolynomialFeatures(degree=2)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)

poly_model = LinearRegression()
poly_model.fit(X_train_poly, y_train)
poly_score = poly_model.score(X_test_poly, y_test)

print("Linear Regression R2 Score:", linear_score)
print("Polynomial Regression R2 Score:", poly_score)
```

```
Linear Regression R2 Score: 0.4526027629719195
Polynomial Regression R2 Score: 0.4156399336408013
```

Result

The **Linear Regression** and **Polynomial Regression** models were successfully implemented and compared using Python. Their performances were evaluated using the **R^2 score**. For this dataset, **Linear Regression** achieved a slightly higher R^2 score than **Polynomial Regression**, indicating better performance.